# Math 54 — Week 3

## Solutions

**P1.**

**Idea.** Independent exactly when there is a pivot in every column.

$R_2 - 3R_1$ gives $(0, 1, -1)$; $R_3 + 2R_1$ gives $(0, 1, 1)$; $R_3 - R_2$ gives $(0, 0, 2)$. Echelon form

$$\begin{bmatrix} 1 & -3 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}:$$

three pivots, so the columns are independent.

**P2.**

**Idea.** Row reduce with $h$ in place; dependence needs a free variable.

$$\begin{bmatrix} 1 & 3 & -1 \\ -1 & -5 & 5 \\ 4 & 7 & h \end{bmatrix} \xrightarrow[R_3-4R_1]{R_2+R_1} \begin{bmatrix} 1 & 3 & -1 \\ 0 & -2 & 4 \\ 0 & -5 & h+4 \end{bmatrix} \xrightarrow{R_3-\frac{5}{2}R_2} \begin{bmatrix} 1 & 3 & -1 \\ 0 & -2 & 4 \\ 0 & 0 & h-6 \end{bmatrix}$$

Dependent exactly when $h = 6$. Then $x_3$ is free; with $x_3 = 1$: $-2x_2 + 4 = 0$ gives $x_2 = 2$, and $x_1 = -3(2) + 1 = -5$. Relation: $-5(1, -1, 4) + 2(3, -5, 7) + (-1, 5, 6) = (0, 0, 0)$. Check: $(-5 + 6 - 1,\ 5 - 10 + 5,\ -20 + 14 + 6)$.

**P3.** (a) Dependent: three vectors in $\mathbb{R}^2$. (b) Dependent: the first is $-2$ times the second. (c) Dependent: contains $\mathbf{0}$. (d) Independent: two vectors, neither a multiple of the other (a multiple of $(1, 2, 0)$ has third entry 0).

**P4.**

**Idea.** Follow $\mathbf{e}_1$ and $\mathbf{e}_2$ through both moves.

$\mathbf{e}_1 = (1, 0)$: reflect $\to (1, 0)$, rotate $\to (0, 1)$. $\mathbf{e}_2 = (0, 1)$: reflect $\to (0, -1)$, rotate $\to (1, 0)$. Standard matrix

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}:$$

$T(x_1, x_2) = (x_2, x_1)$, the reflection through the line $x_2 = x_1$.

**P5.**

$$\left[\begin{array}{rrr|r} 1 & -2 & 3 & -1 \\ 0 & 1 & -1 & 2 \\ 2 & -3 & 5 & 0 \end{array}\right] \to \left[\begin{array}{rrr|r} 1 & 0 & 1 & 3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

($R_3 - 2R_1$ gives $(0, 1, -1, 2)$, equal to $R_2$; then $R_1 + 2R_2$.) $\mathbf{x} = (3, 2, 0) + x_3(-1, 1, 1)$. The coefficient matrix has a free variable (no pivot in column 3): not one-to-one; and only two pivots for three rows: not onto.

**P6.** $T(0,0) = (0,-1) \neq \mathbf{0}$, so $T$ is not linear.

$$S(\mathbf{x}) = \begin{bmatrix} 0 & 1 \\ 3 & -1 \\ 0 & 0 \end{bmatrix} \mathbf{x}$$

is a matrix transformation, so it is linear.

**P7.** The reduced echelon form is

$$\begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

A pivot in every row: onto $\mathbb{R}^3$. Column 4 has no pivot: not one-to-one (as expected, since $4 > 3$).

**P8.**

$$T(3,-2) = T(3\mathbf{e}_1 - 2\mathbf{e}_2) = 3(2,-1,3) - 2(0,4,1) = (6,-11,7).$$

Standard matrix

$$\begin{bmatrix} 2 & 0 \\ -1 & 4 \\ 3 & 1 \end{bmatrix}.$$